

NOTES:

1. REFER TO RUNOFF REDUCTION VOLUME CALCULATIONS

2. REQUIRED DETENTION STORAGE VOLUME = 10,000 CUBIC FEET PER DISTURBED ACRE - REQUIRED RUNOFF REDUCTION VOLUME

POST DEVELOPMENT BASIN #3

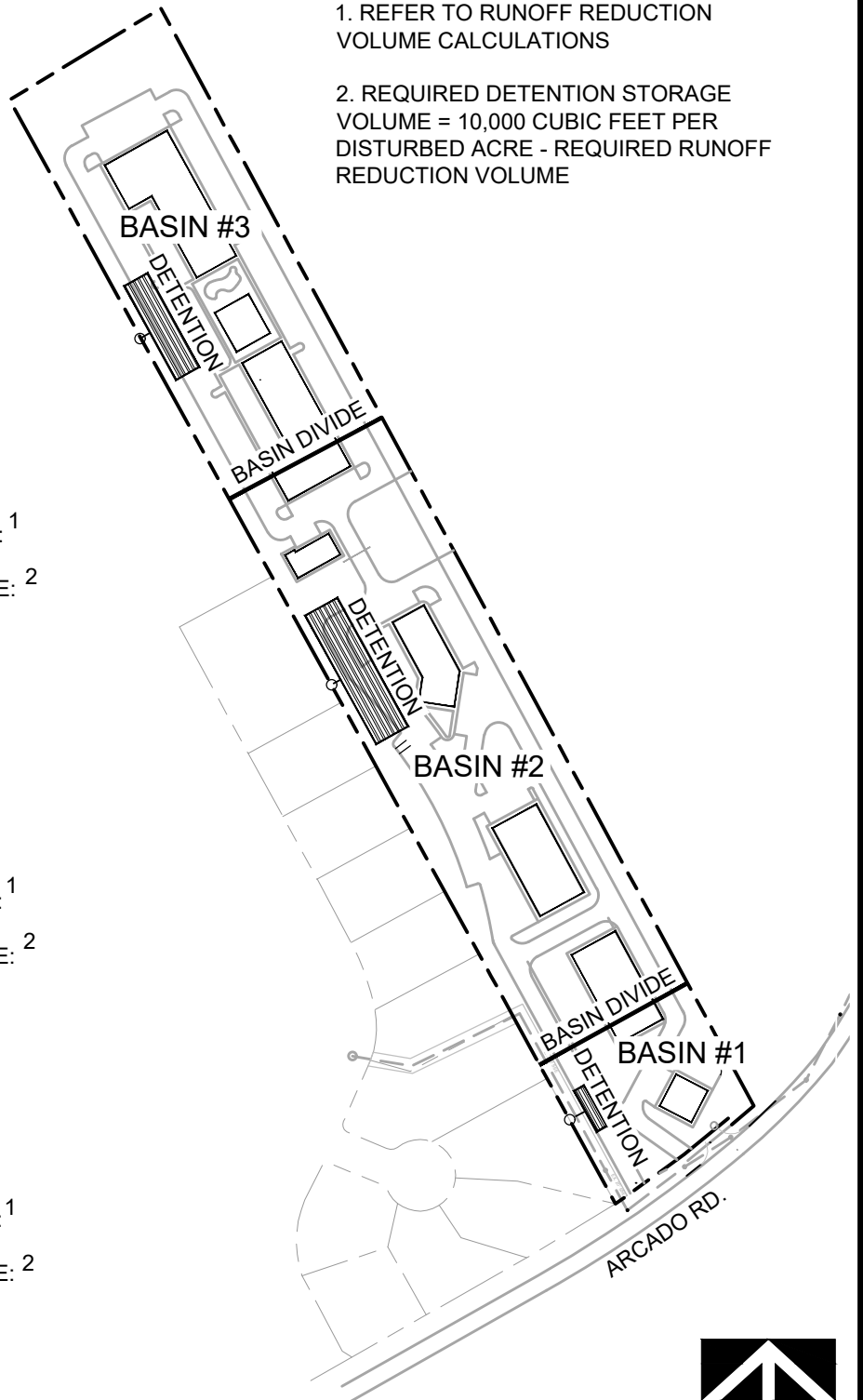
TOTAL AREA: 157642 SF (3.619 AC)
 IMPERVIOUS AREA: 87075 SF (2.00 AC)
 PERVIOUS AREA: 70567 SF
 % IMPERVIOUS: 55%
 REQUIRED RUNOFF REDUCTION VOLUME: ¹
 8,629 CUBIC FEET
 REQUIRED DETENTION STORAGE VOLUME: ²
 27,561 CUBIC FEET

POST DEVELOPMENT BASIN #2

TOTAL AREA: 215,455 SF (4.946 AC)
 IMPERVIOUS AREA: 122,543 SF (2.813 AC)
 PERVIOUS AREA: 92,912 SF
 % IMPERVIOUS: 57%
 REQUIRED RUNOFF REDUCTION VOLUME: ¹
 12,094 CUBIC FEET
 REQUIRED DETENTION STORAGE VOLUME: ²
 37,366 CUBIC FEET

POST DEVELOPMENT BASIN #1

TOTAL AREA: 51,451 SF (1.181 AC)
 IMPERVIOUS AREA: 25,770 SF (0.592 AC)
 PERVIOUS AREA: 25,681 SF
 % IMPERVIOUS: 50%
 REQUIRED RUNOFF REDUCTION VOLUME: ¹
 2,570 CUBIC FEET
 REQUIRED DETENTION STORAGE VOLUME: ²
 9,240 CUBIC FEET



Parker Land Design LLC		Planning	Civil/Site	Landscape
PROJECT	PLD PROJECT NO.	25020	SHEET TITLE	DWG NO.
ARCADO ROAD MULTI-USE	DATE	04-29-2025	PRELIMINARY HYDRO ANALYSIS	HYD-1
	SCALE	1"=250'		
	DRAWN BY	RRP		

PROJECT NAME:	ARCADO RD MULTI-USE LILBURN, GA
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RUNOFF REDUCTION VOLUME CALCULATIONS

"Onsite" (Disturbed Areas)	Total Area (acres)	Impervious (CN = 98)	
POST 1 (TO ALL W.Q. UNITS)	1.18	0.59	
TOTAL	1.18	0.59	

TOTAL DEVELOPED AREA (ACRE)	=	1.18	
TOTAL IMPERVIOUS AREA (ACRE)	=	0.59	
PERCENT IMPERVIOUS (%)	=	49.96	

WATER QUALITY VOULUME:

$$WQr = 1.2 * (RV)$$

$$RV = 0.05 + I * 0.009$$

$$WQv = (WQr/12) * A$$

WHERE:

WQr = WATER QUALITY RUNNOFF

Rv = WEIGHTED VOLUMETRIC RUN OFF
COEFICIENT

I = PERCENT IMPERVIOUS

A = ONSITE AREA

WQr = 1.2 * (0.05+(I *0.009))	=	0.60	
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WQv = WQr/12 *43560*A [CUFT]	=	2570	cubic feet
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PROJECT NAME:	ARCADO RD MULTI-USE LILBURN, GA
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RUNOFF REDUCTION VOLUME CALCULATIONS

"Onsite" (Disturbed Areas)	Total Area (acres)	Impervious (CN = 98)	
POST 2 (TO ALL W.Q. UNITS)	4.95	2.81	
TOTAL	4.95	2.81	

TOTAL DEVELOPED AREA (ACRE)	=	4.95	
TOTAL IMPERVIOUS AREA (ACRE)	=	2.81	
PERCENT IMPERVIOUS (%)	=	56.77	

WATER QUALITY VOULUME:

$$WQr = 1.2 * (RV)$$

$$RV = 0.05 + I * 0.009$$

$$WQv = (WQr/12) * A$$

WHERE:

WQr = WATER QUALITY RUNNOFF
 Rv = WEIGHTED VOLUMETRIC RUN OFF
 COEFICIENT
 I = PERCENT IMPERVIOUS
 A = ONSITE AREA

WQr = 1.2 * (0.05+(I *0.009))	=	0.67	
WQv = WQr/12 *43560*A [CUFT]	=	12094	cubic feet

PROJECT NAME:	ARCADO RD MULTI-USE LILBURN, GA
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RUNOFF REDUCTION VOLUME CALCULATIONS

"Onsite" (Disturbed Areas)	Total Area (acres)	Impervious (CN = 98)
POST 3 (TO ALL W.Q. UNITS)	3.62	2.00
TOTAL	3.62	2.00

TOTAL DEVELOPED AREA (ACRE)	=	3.62
TOTAL IMPERVIOUS AREA (ACRE)	=	2.00
PERCENT IMPERVIOUS (%)	=	55.25

WATER QUALITY VOULUME:

$$WQr = 1.2 * (RV)$$

$$RV = 0.05 + I * 0.009$$

$$WQv = (WQr/12) * A$$

WHERE:

WQr = WATER QUALITY RUNNOFF

Rv = WEIGHTED VOLUMETRIC RUN OFF
COEFICIENT

I = PERCENT IMPERVIOUS

A = ONSITE AREA

WQr = 1.2 * (0.05+(I *0.009))	=	0.66
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WQv = WQr/12 *43560*A [CUFT]	=	8629	cubic feet
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